

Global Analytics

Global Analytics aka GAIN credit is a fin-tech company which develops solutions to provide short-term loans to people with low credit score.

Interview process consisted of 3 rounds after the online test:

Round 1: Programming

This round seemed to test whether you know how to approach a problem, code the solution, and optimize. They will do so giving you the code that you did in the online test and ask you to explain the logic and see if you can do something to optimize your solution. Then you will be given some coding questions and ask you to code. I was given the 4 questions below:

1. Count the number of submatrices that can be formed by 1s in a 2d boolean matrix. (Optimized solution of the problem from the online test)
2. Count the number of islands formed by 1s in a 2d matrix of 0s(water) and 1s(land).
3. Given two strings with some characters and '#' denoting a backspace...check if they both result in the same string after the backspaces are done accordingly(solve with and without extra space)
4. Given an array of positive and negative integers. You are allowed to jump from an index i to: $i+1$, $i-1$, and k in one jump where k is any other index in array where $arr[i] == arr[k]$. Find the minimum number of jumps needed to reach the end of the array if you start from the first index.

Round 2: Technical discussion (Projects, programming, technical questions)

1. Explain your machine learning project.
2. How does Linear regression work?
3. What is Mean Squared Error?
4. What is gradient descent?
5. Explain your website's login module code.
6. How will you use sessions and cookies?
7. Classification vs Regression
8. Normalization forms(1-3)
9. ACID properties in RDBMS
10. How will you achieve Isolation in RDBMS?
11. Write a code to find the height of BST.

Round 3: (Talk with Director, Technical + HR)

Given a C program based on pointers, asked to find the output, explain what happens in each line. The program was something like this:

```
#include<stdio.h>
#include<stdlib.h>
#include<string.h>
void func(char *str)
{
    str=(char*)malloc(500);
    strcpy(str,"global analytics");
}
int main(void)
{
    char *str=(char *)malloc(50);
    strcpy(str,"hello");
    func(str);
}
```

```
printf("%s -- %d",str,(int)strlen(str));  
return 0; }
```

1. Some in-depth questions about the memory allocation and the pointers in the program like where are the variables stored actually and how they are stored.
2. What is the biggest number a char can hold?
3. What will be inside stdio.h?
4. What is the use of python's dictionary?
5. Explain any one project.
6. He also picked another project from my GitHub and asked me to explain the code.

Topics to be focussed:

- ◆ Problem solving - Coding & Optimization
- ◆ Thorough knowledge of projects mentioned in resume
- ◆ Core subjects like DBMS, C, data structures.

Take away:

If you made it through the online test, be ready with optimization ideas for the problem you did on the test. Have a thorough knowledge of your projects and subjects you mention in your resume.

Hope this helps.

Thank you and All the Best.

Nithish Kumar R
nithishkumar642000@gmail.com